

PROJECT AND TEAM INFORMATION

Project Title

RescueRoute – Disaster Response and Evacuation Management System

Student / Team Information

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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

Natural disasters can be extremely difficult to manage, especially when roads are blocked or damaged. In such situations, reaching people who need help can take more time. Recent earthquakes and other natural calamities in Nepal have also shown the damage that disasters can cause and the difficulties involved in providing timely assistance.

We have also considered emergency cases where patients need to reach a hospital as quickly as possible. Blocked roads, a limited number of ambulances and hospital capacity can make the response even more difficult.

Therefore, our project focuses on developing a system that can help organize disaster response and evacuation using Data Structures and Object Oriented Programming Concepts.

State of the Art / Current solution (1 pt)

In our project roads will be represented as a graph using an adjacency list. Locations will be nodes and roads will be edges in the graph. When a road is blocked the road will be ignored during the path finding process. BFS will be implemented to find a path, in the road network. Priority queues will be used to handle evacuation and emergency requests according to their priorities. Object-oriented programming will be used to represent patients, ambulances and hospitals.

Project Goals and Milestones (2 pts)

Goal:

The goal of this project is to create a Disaster and Evacuation Management System that is built using C/C++. This system will handle locations, roads, emergency requests, patients, ambulances and hospitals. Graphs and BFS will be used to find routes that're available. Priority Queues will be used to manage emergency requests based on how urgent they're. The system will combine Data Structures and OOP concepts, into one working project.

Milestones:

1. Core Foundation Create the basic system structure of patients, ambulances, hospitals, disasters and evacuees.
2. Road Network Construct the road network as a graph and provide roads with availability / unavailability.
3. Route Finding
Use BFS to find the path.
4. Emergency Priority
Priority Queue processes the critical emergency cases by priority.
5. Record Management
Linked Lists, Stacks, Searching and Sorting manage and organize system records.
6. Disaster Simulation.
Simulate various disaster scenarios that affect roads, patients, emergencies and hospital availability.
7. Testing.
Test the system in various disaster and emergency scenarios.

Project Approach (3 pts)

As this is a combined Project-Based Learning (PBL) assignment, we will build the project using C and C++. Data Structures will be combined with Object Oriented Programming (OOP) to create a disaster and evacuation management system.

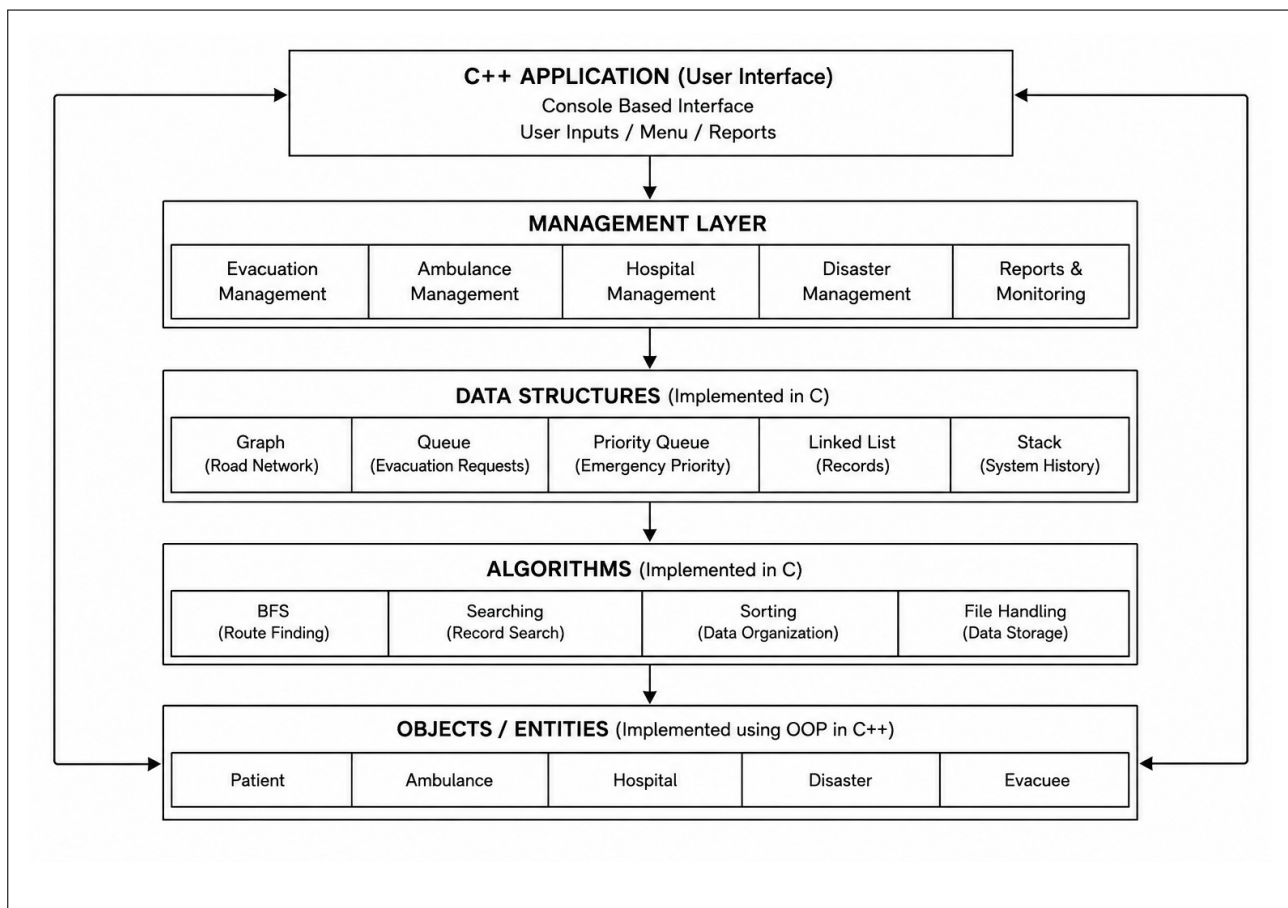
OOP Topics Used:

- 1. Classes and Objects:** We will use classes to represent objects such as patients, ambulances, hospitals, disasters and evacuees. Each object will contain the information related to it.
- 2. Encapsulation:** Information related to patients, hospitals and ambulances will be kept inside their respective classes and accessed through suitable functions.

DS Topics Used:

- 1. Graphs:** Locations will be represented as nodes and roads as connections between them.
- 2. BFS (Breadth-First Search):** BFS will be used to find available routes while avoiding blocked roads.
- 3. Queues:** Queues will be used to manage evacuation and emergency requests in the order in which they are received.
- 4. Priority Queues:** Critical emergency cases will be given higher priority so that they can be handled first.
- 5. Linked Lists:** Linked Lists will be used to manage records that may change during the simulation.
- 6. Stacks:** Stacks will be used to keep track of recent system actions.
- 7. Searching and Sorting:** Searching will help locate required records, while sorting will organize the stored information.
- 8. File Handling:** Files will be used to save important records so that the information can be accessed later.

System Architecture (High Level Diagram)(2 pts)



Project Outcome / Deliverables (1 pts)

This project will result in a working Disaster and Evacuation Management System using Data Structures in C and Object-Oriented Programming in C++.

1. A working system to register disasters, patients, evacuees, hospitals and ambulances.
2. Graphs and BFS for representing locations and finding available routes when roads are blocked.
3. Queues for evacuation requests and Priority Queues for critical emergency cases.
4. Linked Lists, Stacks, Searching and Sorting techniques for record management, system history and organizing data.
5. C++ classes for Patient, Ambulance, Hospital, Disaster and Evacuee.
6. Disaster simulations demonstrating route finding, ambulance dispatch and hospital allocation.
7. File Handling for saving records, along with complete source code and final project demonstration.

Assumptions

We assume that the system will use a simplified, unweighted road graph where each road can have a LOW, MEDIUM, HIGH or BLOCKED status. BFS will be used for optimal paths between locations.

Hospital capacity, ambulance availability and patient severity will be stored as basic records and used during the simulation.

The system is designed as a small-scale academic simulation, so real-world factors such as live GPS data, GIS mapping, real-time traffic conditions and cloud services are not considered. Dijkstra's algorithm is also outside the scope, as BFS is sufficient for the simplified road network used in the project.

References

1. Kazi Ashik Islam et al., "A Simulation-based Approach for Large-scale Evacuation Planning," IEEE, 2020. IEEE Xplore
2. Cormen et al., *Introduction to Algorithms*, MIT Press. MIT Press
3. Bjarne Stroustrup, *The C++ Programming Language*. C++ Programming Language